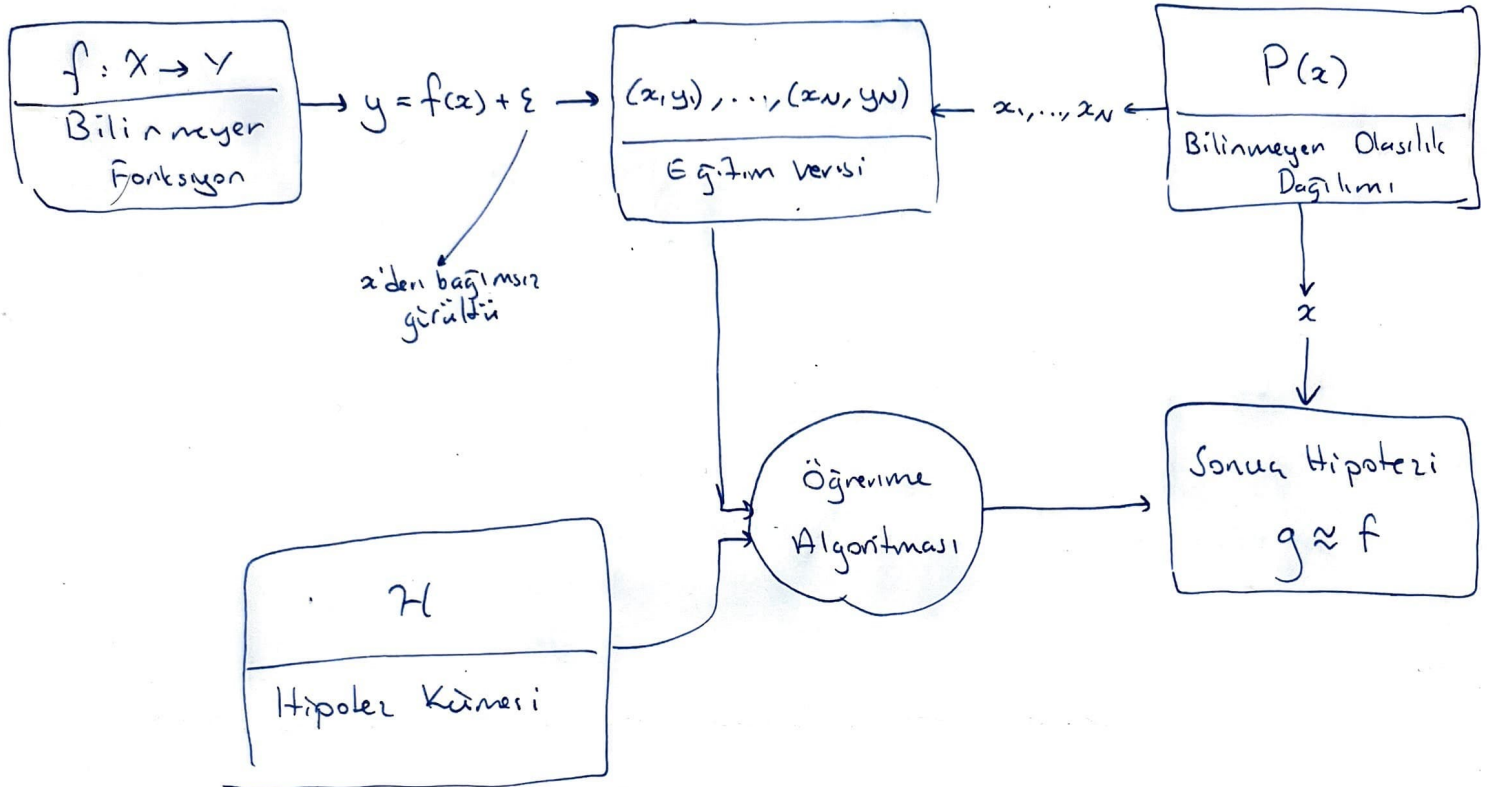


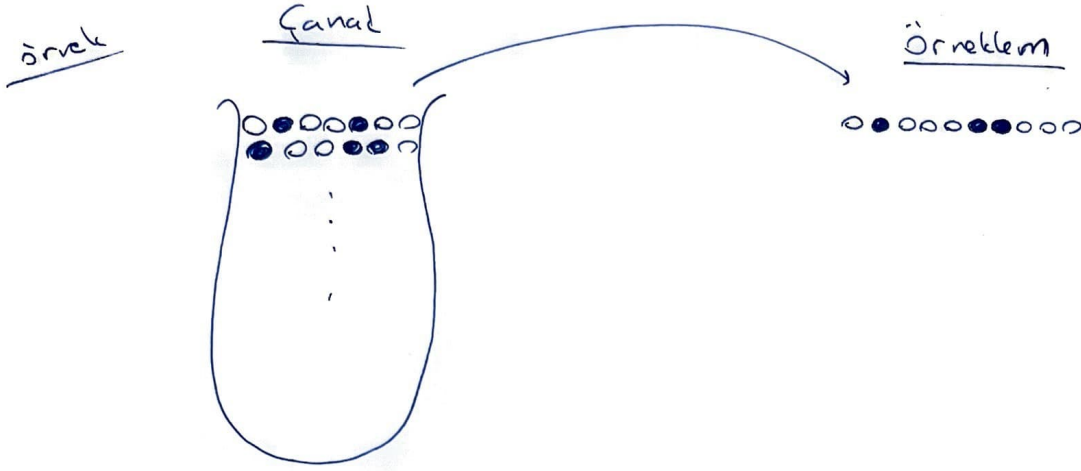
Yapay Öğrenmeye Giriş - NMK

X : Olası bütün Girdiler

y : Olası bütün Çıktılar



x	y	g	f_1	f_2	f_3	$\dots$	f_8
000	0	0	0	0	0		0
001	1	1	1	1	1		1
010	1	1	1	1	1	...	1
011	0	0	0	0	0		0
100	1	1	1	1	1		1
101		?	0	0	0		1
110		?	0	0	1	...	1
111		?	0	1	0		0



μ : siyah gelme olasılığı

ν = örneklemedeki siyah oranı

~~mu = ?~~

Soru: $\mu = 0.9$ ise $\nu = 0.1$ olma olasılığı nedir? $N = 10$ için.

$$P(X=1) = \binom{10}{1} \mu^1 \cdot (1-\mu)^9 \quad (X = X_1 + \dots + X_N)$$

Binomial Dağılım

$$= 10 \cdot 0.9 \cdot (0.1)^9 = \frac{9}{10^9}$$

$$P(X=0) = (1-\mu)^{10} = \frac{1}{10^{10}}$$

cevap = $\frac{9}{10^9} + \frac{1}{10^{10}} = \frac{91}{10^{10}}$

$$\nu \rightarrow \mu = ?$$

Hoeffding Eşitsizliği

Örnekleme büyüklüğü N ise her $\epsilon > 0$ için

$$P[|\nu - \mu| > \epsilon] \leq 2e^{-2\epsilon^2 N}$$

Hoeffding's Inequality

Theorem let $Z_1, \dots, Z_n$ be independent random variables on $\mathbb{R}$ s.t. $a_i \leq Z_i \leq b_i$ with probability one. If $S_n = \sum_{i=1}^n Z_i$, then for all $t > 0$

$$P[S_n - \mathbb{E}[S_n] \geq t] \leq e^{-2t^2 / \sum (b_i - a_i)^2}$$

and

$$P[S_n - \mathbb{E}[S_n] \leq -t] \leq e^{-2t^2 / \sum (b_i - a_i)^2}.$$

Remark 1: $P[|S_n - \mathbb{E}[S_n]| \geq t] \leq 2e^{-2t^2 / \sum (b_i - a_i)^2}.$

Remark 2: $Z_i \stackrel{i.i.d.}{\sim} \text{Ber}(p) \Rightarrow a_i = 0, b_i = 1$ and $S_n \sim \text{binom}(n, p).$

Thus $\mathbb{E}[S_n] = np$, and we have

$$\begin{aligned} P[|S_n - np| \geq t] &\leq 2e^{-2t^2/n} \\ \Rightarrow P[|S_n/n - p| \geq t/n] &\leq 2e^{-2t^2/n \cdot n} \\ \Rightarrow \underbrace{P[|S_n/n - p| \geq \varepsilon]}_{\substack{\nu \\ \mu}} &\leq 2e^{-2\varepsilon^2 \cdot n} \end{aligned}$$

Hoeffding Inequality: let sample size be N . Then for ~~the~~ all $\varepsilon > 0$

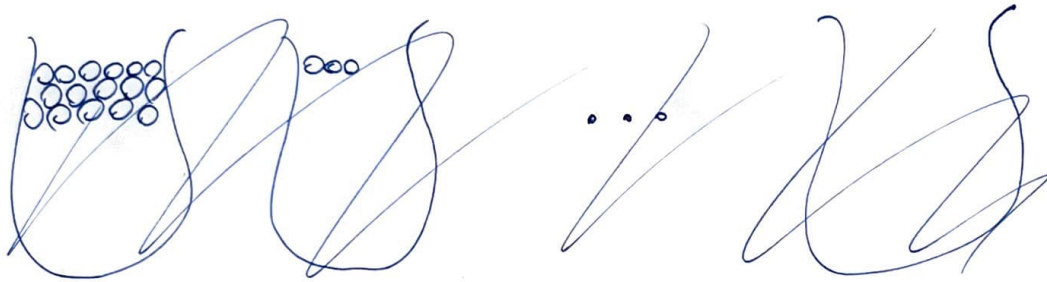
$$P[|\nu - \mu| \geq \varepsilon] \leq 2e^{-2\varepsilon^2 N}$$

Soru Yukarıdaki soruyu Hoeffding Eşitsizliği ile çözünüz.

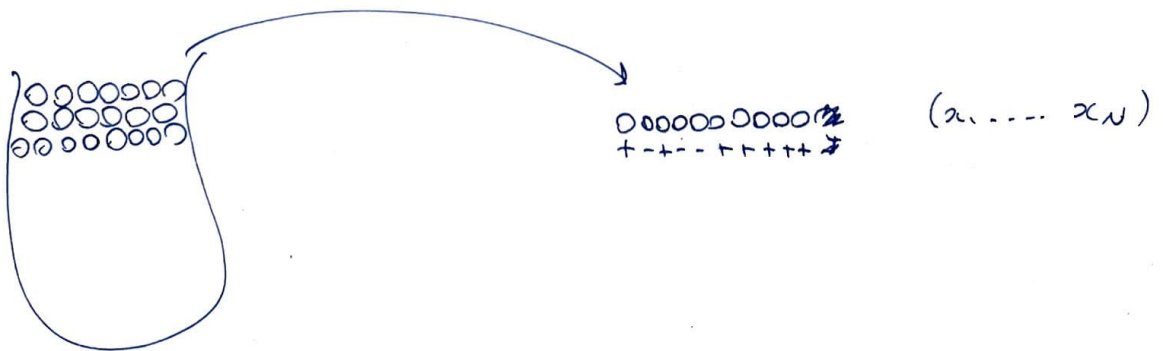
$$|V - \mu| = |0.1 - 0.9| = 0.8$$

$$P[|V - \mu| \geq 0.8] \leq 2 e^{-2 \cdot 10 \cdot 0.8^2} = 2 e^{-12.8} \approx 5.52 \times 10^{-6}$$

Peki $\mathcal{H} = \{h_1, \dots, h_M\}$ ise



Peki elimizde $h \in \mathcal{H}$ varsa



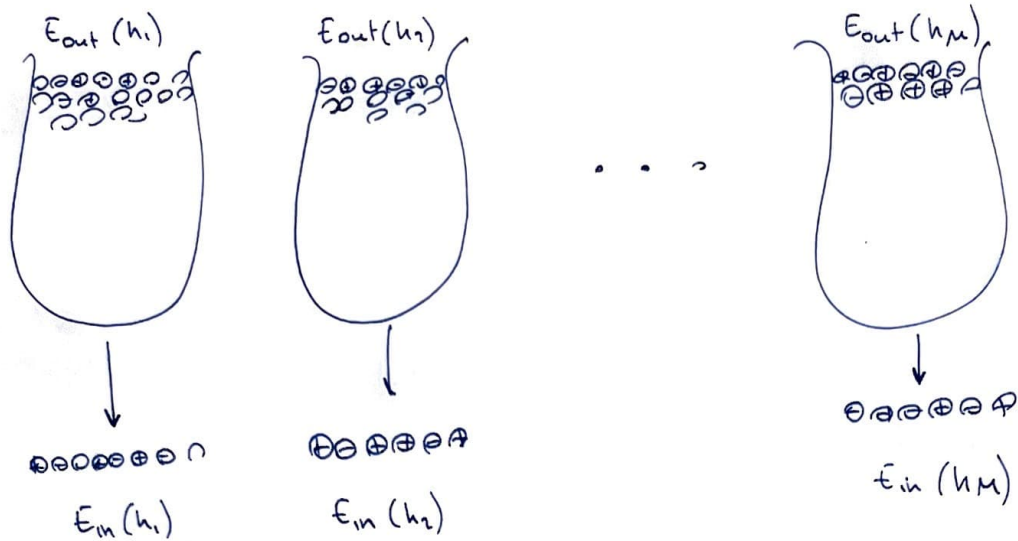
$$E_{out}(h) = P[h(x) \neq f(x)] = \sum_{i=1}^N \mathbb{1}[h(x_i) \neq f(x_i)]$$

$$E_{in}(h) = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[h(x_i) \neq f(x_i)] = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[h(x_i) \neq f(x_i)]$$

$\mathbb{1}(p) = \begin{cases} 1 & \text{if } p \text{ is true} \\ 0 & \text{if } p \text{ is false} \end{cases}$

$$\Rightarrow P[|E_{in}(h) - E_{out}(h)| \geq \epsilon] \leq 2 e^{-2 \epsilon^2 N}, \quad \forall \epsilon > 0.$$

Pelki $H = \{h_1, \dots, h_M\}$ ise?



$$g \in H \text{ ise } P[|E_{in}(g) - E_{out}(g)| > \epsilon] = ?$$

$$\underbrace{P_0}_{P_0} \Rightarrow \underbrace{P_1}_{P_1} \vee \underbrace{P_2}_{P_2} \vee \dots \vee \underbrace{P_M}_{P_M}$$

$$P_0 \Rightarrow P_1 \vee \dots \vee P_M \Rightarrow P(P_0) \leq P(P_1 \vee \dots \vee P_M) \leq P(P_1) + \dots + P(P_M)$$

$$\begin{aligned} \text{yani } P[|E_{in}(g) - E_{out}(g)| > \epsilon] &\leq \sum_{i=1}^M P[|E_{in}(h_i) - E_{out}(h_i)| > \epsilon] \\ &\leq 2M e^{-2\epsilon^2 N} \end{aligned}$$

Genelleme Hatası

$$P[|E_{in}(g) - E_{out}(g)| > \epsilon] \leq 2M e^{-2\epsilon^2 N}, \quad \forall \epsilon > 0.$$

$$2M e^{-2\epsilon^2 N} = \delta \quad \Rightarrow \quad \frac{2M}{\delta} = e^{+2\epsilon^2 N} \quad \Rightarrow \quad +2\epsilon^2 N = \ln \frac{2M}{\delta}$$

$$\Rightarrow \epsilon = \sqrt{\frac{1}{2N} \ln \frac{2M}{\delta}}$$

$$\Rightarrow P\left[|E_{in}(g) - E_{out}(g)| > \sqrt{\frac{1}{2N} \ln \frac{2M}{\delta}}\right] \leq \delta$$

$$\Rightarrow 1 - \delta \text{ olasılık ile } |E_{in}(g) - E_{out}(g)| \leq \sqrt{\frac{1}{2N} \ln \frac{2M}{\delta}}$$

$$\Rightarrow 1 - \delta \text{ olasılık ile } \underbrace{E_{in}(g) - \sqrt{\frac{1}{2N} \ln \left(\frac{2M}{\delta}\right)}}_{(1)} \leq E_{out}(g) \leq \underbrace{E_{in}(g) + \sqrt{\frac{1}{2N} \ln \left(\frac{2M}{\delta}\right)}}_{(2)}$$

(2) $\rightarrow$ maks. hata

(1) $\rightarrow$ Ne kadar
Daha iyi g seçebiliyoruz? ~~Ne kadar~~

VC - Boyutu (Vapnik - Chervonenkis)

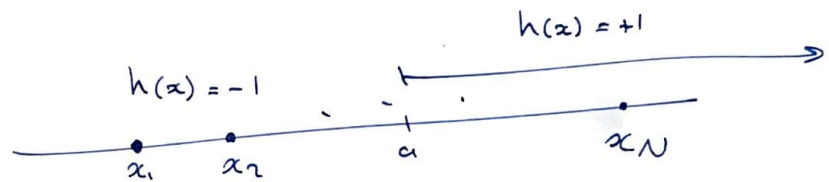
$$h \in \mathcal{H} \text{ ve } x \in \mathcal{X} \text{ ise } h(x) \in \{-1, 1\}$$

Tanım: $x_1, \dots, x_N \in \mathcal{X} \Rightarrow \mathcal{H}(x_1, \dots, x_N) = \{ (h(x_1), \dots, h(x_N)) : h \in \mathcal{H} \}$

Tanım: $m_{\mathcal{H}}(N) = \max_{x_1, \dots, x_N \in \mathcal{X}} |\mathcal{H}(x_1, \dots, x_N)|$. $(|A| : A \text{ kümesinin eleman sayısı})$

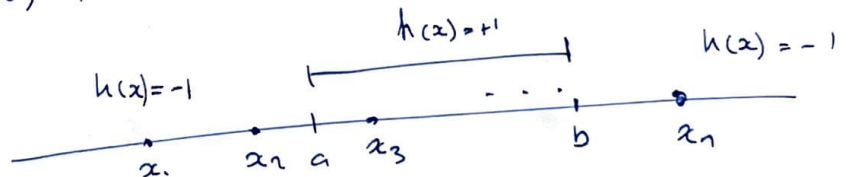
örnek

① pozitif ışınlar



$$m_{\mathcal{H}}(N) = N+1$$

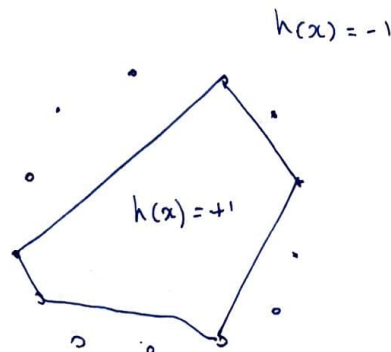
② pozitif aralıklar



$$m_{\mathcal{H}}(N) = \binom{N+1}{2} + 1 = \frac{1}{2} N^2 + \frac{1}{2} N + 1$$

$$\frac{(N+1)!}{2! (N-1)!}$$

③ Konveks Wörder $\mathbb{R}^2$ 'de



Teorem: $m_H(k) < 2^k \Rightarrow m_H(N) \leq \sum_{i=0}^{k-1} \binom{N}{i}, \forall N$

$\downarrow$
kırılma noktası

Yani $m_H(N)$ N 'ye göre $(k-1)$. dereceden polinom.

Tanım: $d_{vc}(H) \neq \infty$ $m_H(N) = 2^N$ koşulunu sağlayan en büyük N . tam sayısı.

Örneğin $k = d_{vc} + 1$ kırılma noktası.

$d_{vc} < \infty$ ise

$$m_H(N) \leq \sum_{i=0}^{d_{vc}} \binom{N}{i} \leq N^{d_{vc}} + 1$$

Teorem: Her $\delta > 0$ için $1-\delta$ olasılıkla

$$E_{out}(g) \leq E_{in}(g) + \sqrt{\frac{8}{N} \ln \left(\frac{4 m_H(2N)}{\delta} \right)}$$

Proof of Hoeffding Inequality

Markov's Inequality: If $X \geq 0$, then $P(X \geq a) \leq \frac{E(X)}{a}$.

Proof:

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f(x) dx = \int_0^{\infty} x f(x) dx = \int_0^a x f(x) dx + \int_a^{\infty} x f(x) dx \\ &\quad \downarrow \\ &\quad X \geq 0 \\ &\geq \int_a^{\infty} x f(x) dx \\ &\geq \int_a^{\infty} a f(x) dx = a \int_a^{\infty} f(x) dx \\ &= a P(X \geq a). \end{aligned}$$

Thus $P(X \geq a) \leq \frac{E(X)}{a}$.

Hoeffding's lemma: let X be any real-valued r.v. st $a \leq X \leq b$ a.s. (i.e. with probability one). Then, for all $\lambda \in \mathbb{R}$

$$E(e^{\lambda X}) \leq e^{\left(\lambda E(X) + \frac{\lambda^2 (b-a)^2}{8}\right)},$$

or equivalently

$$E\left[e^{\lambda(X - E(X))}\right] \leq e^{\frac{\lambda^2 (b-a)^2}{8}}.$$

Proof:

Let $\mu = \mathbb{E}(X)$. WLOG, replace X by $X - \mu$, a by $a - \mu$, b by $b - \mu$.
and assume that $\mathbb{E}[X] = 0$ and $a \leq 0 \leq b$.

Since $e^{\lambda x}$ is convex $\left(\text{ie for } f \text{ is convex iff for } 0 \leq t \leq 1 \text{ and } x_1, x_2 \in X \right)$

$$f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2)$$

for all $x \in [a, b]$ and $t = \frac{b-x}{b-a} \in [0, 1]$. Then,

$$ta + (1-t)b = \frac{b-x}{b-a} a + \frac{x-a}{b-a} b = \frac{ab - xa + xb - ab}{b-a} = \frac{x(b-a)}{b-a} = x.$$

Therefore,

$$e^{\lambda x} \leq \frac{b-x}{b-a} e^{\lambda a} + \frac{x-a}{b-a} e^{\lambda b}.$$

So

$$\begin{aligned} \mathbb{E}[e^{\lambda X}] &= \frac{b - \overset{0}{\mathbb{E}(X)}}{b-a} e^{\lambda a} + \frac{\overset{0}{\mathbb{E}(X)} - a}{b-a} e^{\lambda b} \\ &= \frac{b}{b-a} e^{\lambda a} + \frac{-a}{b-a} e^{\lambda b} \\ &= e^{L(\lambda(b-a))} \end{aligned}$$

where $L(h) = \frac{ha}{b-a} + \ln \left(1 + \frac{a - e^{ha}}{b-a} \right).$

$$L(0) = L'(0) = 0 \quad \text{and} \quad L''(h) = - \frac{abeh}{(b - aeh)^2} \stackrel{a \leq 0}{\leq} \frac{abeh}{4a^2 e^h} = \frac{1}{4}.$$

(*)

(*) $0 \leq (x-y)^2 = x^2 + y^2 - 2xy = x^2 + y^2 + 2xy - 4xy = (x+y)^2 - 4xy$
 $\Rightarrow (x+y)^2 \geq 4xy \quad \left(\text{for } x, y \geq 0 \right)$

Thus, from Taylor's Theorem, $\exists \theta \in [a, b]$ s.t.

$$L(h) = L(0) + h L'(0) + \frac{1}{2} h^2 L''(\theta) \leq \frac{h^2}{8}.$$

$$\text{Thus, } \mathbb{E}[e^{\lambda x}] \leq e^{\frac{\lambda^2(b-a)^2}{8}}.$$

Hoeffding's Inequality: Let $X_1, \dots, X_n$ be independent random variables s.t. $a_i \leq X_i \leq b_i$ a.s. Consider $S_n = X_1 + \dots + X_n$.

Then, for all $t > 0$

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq t) \leq e^{-2t^2 / \sum_{i=1}^n (b_i - a_i)^2}$$

and

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \leq -t) \leq e^{-2t^2 / \sum_{i=1}^n (b_i - a_i)^2}$$

Proof: $\mathbb{P}(S_n - \mathbb{E}[S_n] \geq t) = \mathbb{P}(e^{s(S_n - \mathbb{E}[S_n])} \geq e^{st})$

Markov's Ineq. $\leftarrow \leq e^{-st} \mathbb{E}[e^{s(S_n - \mathbb{E}[S_n])}]$

$S_n = X_1 + \dots + X_n \leftarrow = e^{-st} \prod_{i=1}^n \mathbb{E}[e^{s(X_i - \mathbb{E}[X_i])}]$

Hoeffding's lemma $\leftarrow \leq e^{-st} \prod_{i=1}^n e^{\frac{s^2(b_i - a_i)^2}{8}}$

$$= e^{\underbrace{\left(-st + \frac{1}{8} s^2 \sum_{i=1}^n (b_i - a_i)^2\right)}_{\text{gets its max at } s = \frac{4t}{\sum_{i=1}^n (b_i - a_i)^2}}}$$

Putting this s in the quadratic function: $\mathbb{P}[S_n - \mathbb{E}(S_n) \geq t] \leq e^{-2t^2 / \sum (b_i - a_i)^2}.$

Definition: $B(N, k)$ is the max. number of dichotomies on N points s.t. no subset of size k of the N points can be shattered by these dichotomies.

Note: If k is a break point for $\mathcal{H}$ then $m_{\mathcal{H}}(N) \leq B(N, k)$.

- $B(N, 1) = 1$ for all N .
- $B(1, k) = 2$ for all $k \geq 1$.

Now assume $N \geq 2$ and $k \geq 2$.

	# of rows	x_1	x_2	$\dots$	x_{N-1}	x_N
S_1	α	+1	+1	$\dots$	+1	+1
		-1	+1	$\dots$	+1	-1
		$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
		+1	-1	$\dots$	-1	-1
		-1	+1	$\dots$	-1	+1
S_2^+	β	+1	-1	$\dots$	+1	+1
		-1	-1	$\dots$	+1	+1
		$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
		+1	-1	$\dots$	+1	+1
		-1	-1	$\dots$	-1	+1
S_2^-	β	+1	-1	$\dots$	+1	-1
		-1	-1	$\dots$	+1	-1
		$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
		+1	-1	$\dots$	+1	-1
		-1	-1	$\dots$	-1	-1

$$B(N, k) = \alpha + 2\beta$$

Since $S_2^+ = S_2^-$ we have $\alpha + \beta \leq B(N-1, k)$

Also no subset of size $k-1$ can be shattered by S_2^+ otherwise $S_2^+ \cup S_2^-$ would shatter k points when x_N added. Thus

$$\beta \leq B(N-1, k-1)$$

Therefore $B(N, k) \leq B(N-1, k) + B(N-1, k-1)$

		k						
		1	2	3	4	5	6	...
N	1	1	2	2	2	2	2	...
	2	1	3	4	4	4	4	...
	3	1	4	7	8	8	8	...
	4	1	5	11	15	16	16	...
	5	1	6	:	...	...	...	
	6	1	7	:	...	...	...	
	:	:	:	:	...	...	...	

Sauer's Lemma: $B(N, k) \leq \sum_{i=0}^{k-1} \binom{N}{i}.$

Proof: For $N=1, k=1$ we have $B(1, 1) = 1 = \sum_{i=0}^0 \binom{1}{i} = \binom{1}{0} = 1.$

The proof is by induction on N .

Assume the statement is true for all $N \leq N_0$ and all k .

We need to prove the statement for $N=N_0+1$ and all k .

Since the statement is true for $k=1$ ($B(N, 1) = 1$), we assume $k \geq 2$.

$$B(N_0+1, k) \leq B(N_0, k) + B(N_0, k-1)$$

$$\leq \sum_{i=0}^{k-1} \binom{N_0}{i} + \sum_{i=0}^{k-2} \binom{N_0}{i}$$

$$= 1 + \sum_{i=1}^{k-1} \binom{N_0}{i} + \sum_{i=1}^{k-1} \binom{N_0}{i-1}$$

$$= 1 + \sum_{i=1}^{k-1} \left[\binom{N_0}{i} + \binom{N_0}{i-1} \right]$$

$$= 1 + \sum_{i=1}^{k-1} \binom{N_0+1}{i}$$

$$= \sum_{i=0}^{k-1} \binom{N_0+1}{i}$$

Theorem: If $m_{\mathcal{H}}(k) < 2^k$ for some k , then $m_{\mathcal{H}}(N) \leq \sum_{i=0}^{k-1} \binom{N}{i}$

for all N .